

Enhanced antifouling performance with a synergistic effect of self-renewal antifouling and contact bacteriostasis via the ionic exchange

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ABSTRACT

It is extremely difficult for traditional coatings for antifouling constructed around a single antifouling strategy to adapting to the demands of complex marine environment. Herein, a poly-ionic liquid-based germicidal polyacrylate-3 coating (AFC-3) was prepared via free radical photo-copolymerization using isooctyl acrylate (2-EHA), butyl acrylate (BA) and ionic liquid monomer (IL-PF₆) as monomers. Besides the contact bacteriostasis mechanisms, the introduction of IL-PF₆ achieved the switch between self-renewal and hydrophilic surface by the exchange of PF₆⁻ and Cl⁻ in artificial seawater. Regarding *E. coli* and *S. aureus*, the AFC-3 coating showed antibacterial efficiency of 100 % and 99.96 %, respectively. Moreover, with the continuous ionic exchange between PF₆⁻ and Cl⁻, the angle towards the water contact (WAC) of the coating became 11.356 °C, contributing to the enhancement of anti-protein property on the surface of the coating. The AFC-3 coating thus exhibited significantly effective antifouling performances, and is a highly promising coating for a broader range of marine antifouling and biomedical materials. Besides, the transmittance of the coating was above 92 % between 400 and 800 nm⁻¹, conferring it great potentials for applications on the surface of optical devices like underwater cameras.

1. Introduction

Marine biofouling accumulation on ships and offshore drilling platforms causes huge economic losses and damages to the ecosystem, and is therefore the most significant threat to the shipping industry [1–8]. Recently, the use of coatings has been acknowledged as one of the most practical and economical techniques for the prevention of marine biological fouling adhesion [9]. Although anti-fouling coatings containing toxic metals and fungicides are known to be cheap in addition to their good bactericidal effect, they pose great harm to the human health and marine environment. Therefore, the development of efficient antifouling coatings has attracted increasing attention. Acrylate copolymers, polyurethanes and polydimethylsiloxanes are commonly used as resin matrix in antifouling coatings. Compared with polyurethanes and polydimethylsiloxanes, acrylic resin coatings exhibit better film

formation, increased weather resistance, higher transparency, stronger adhesion to the substrate, and are relatively cheaper. However, their widespread application is limited by their poor thermal stability, sticky nature at room temperature, and high surface energy. Hence, it is necessary to modify acrylic resins to be able to meet basic requirements for anti-fouling coatings [10–12].

Many antifouling polymers with antifouling groups have been reported in recent times. On the one hand, antifouling coatings based on hydrophilic and zwitterionic polymers [13,14], such as poly(ethylene glycol) (PEG) [15], poly(acrylic acid) (PAA) [15], poly(acrylamide) (PAAm) [16], poly(sulfobetaine methacrylate) (PSBMA), and poly(carboxybetaine methacrylate) (PCBMA), primarily rely on the generation of hydrated barriers that inhibit bacterium and other microbiological particles from attaching without actually killing the organisms; on the other hand, literatures contain reports of several bactericides that have

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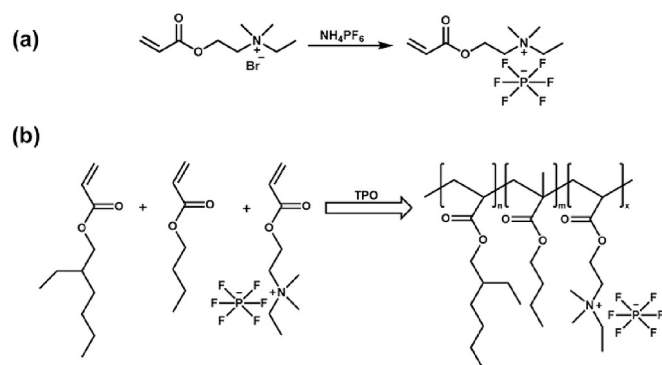


Fig. 1. (a) Synthesis routes of the IL-PF₆ and (b) the AFC-x coatings.

been incorporated into antimicrobial coatings, such as antibiotics [17], silver metal and salts [18], nitric oxide [19], and antimicrobial peptides [20]. However, drawbacks like their apparent cytotoxicity, narrow antimicrobial spectrum, and implications for propagating multidrug resistance, have limited their widespread application.

A good deal of anti-bacterial coatings employ compounds containing quaternary ammonium (QACs) to account for their broad-spectrum bactericidal abilities, which promotes the killing of bacteria and other microbial while being harmless to the ecological environment [21]. The dead bacteria would no longer be able to reproduce on the surfaces of coatings, thus sliding down with the impact of the water. In cases where the dead bacteria are left accumulated on the surface of the coatings, they decrease the effects of QACs by impeding their contact with live bacteria. Therefore, QACs alone cannot achieve the ideal antifouling ability, and the fabrication of self-renewing coating surfaces is of great importance for the release of fouling.

Herein, an enhanced antifouling strategy for switching between self-renewing and hydrophilic surfaces was fabricated by the exchange of PF₆⁻ and Cl⁻ in artificial seawater and contact bactericidal mechanisms. The self-renewing surface contributing to fouling release and the formation of hydrated layers is also conducive for the resistance of protein adhesion. As a result, the 2-EHA, BA, and IL-PF₆ obtained by an ion exchange reaction were used to synthesize the AFC-3 coating via the free radical photo-copolymerization. It possessed a potent synergistic consequence of three antifouling methodologies, exhibiting significant effective antifouling value. Moreover, the AFC-3 coating showed potentials for applications on the surface of optical devices, such as underwater cameras, due to its excellent light transmittance.

2. Experimental section

2.1. Materials

Ammonium hexafluorophosphate (AHFP, 98 %) was purchased from Maclin. Butyl acrylate (BA, 99 %) and *N,N*-dimethylformamide (DMF, 98 %) were attained from Aladdin. Isooctyl acrylate (2-EHA, 99 %) and diphenyl - (2, 4, 6-trimethylbenzoyl) phosphorus oxide (TPO, 98 %) were received from Zhengzhou Alpha Chemical Co., LTD. Bacteria-cellulose emulsion (BC, 0.8 %) was supplied by Guilin Qihong technology Co., Ltd. Bromoethane and ethyl acetate were supplied by Sino-pharm Group Chemical Reagent Co., Ltd. Without additional purification, all of the reagents were utilised. Throughout the experimentation, deionized water was used.

2.2. Preparation of the IL-PF₆

The hydrophilic ionic liquid monomer (IL-Br) was attained using established methods described in the literature [22]. Firstly, IL-Br (10 g, 39.7 mmol) and AHFP (6.47 g, 39.7 mmol) were used for the complete dissolution to a saturated state. The AHFP solution was then combined

with the aforementioned IL-Br solution, which was then agitated for 12 h at room temperature to allow for the precipitation of white crystals. Finally, the white crystals were then freeze-dried in a vacuum freeze-dryer for 72 h to produce IL-PF₆ after being rinsed three times with deionized water. (Fig. 1a).

2.3. Preparation of ionic liquid based poly-acrylate-*x* (AFC-*x*) coatings

A certain amount of 2-EHA, BA and IL-PF₆ was initially dissolved in quantified DMF, followed by the addition of TPO as a photo-initiator. Via free radical photo-polymerization, AFC-*x* coatings were synthesized under ultraviolet light for 30 min at a wavelength of 405 nm (Fig. 1b). The specific contents of the added reagents are listed in Table S1.

Preparation method of LB (Luria-Bertani) medium: 200 mL of deionized water was stirred in 2 g of sodium chloride, 2 g of tryptone, and 1 g of yeast extract powder before sterilization by high pressure steam for 20 min at 120 °C.

Preparation method of Solid LB medium: An identical procedure to that of the Preparation method of LB (Luria-Bertani) medium was used to prepare the Solid LB medium. 200 mL of deionized water was stirred in 2 g of sodium chloride, 2 g of tryptone, 1 g of yeast extract powder and 4 g agar powder before sterilization by a high-pressure steam for 20 min at 120 °C.

Preparation method of bacterial culture medium: The strains were inoculated into the liquid LB medium and cultured in a micro-oscillating incubator at 310 K for 24 h. The optical density (OD) of the bacterial culture medium was 1.

Firstly, the test sample was submerged in the bacterial culture medium and cultured at 37 °C for 24 h, and then taken out. Next, the surface of the sample was gently washed with ultra-pure water to flush the loosely attached bacteria. Then the sample was transferred into 20 mL PBS buffer for ultrasound for 5 min to elute the adhered bacteria on the surface into the 20 mL PBS buffer. Besides the above-mentioned 20 mL PBS buffer mentioned above was diluted 1000 times, and 25 μ L of the diluent was applied evenly on different solid media. Finally, the different solid media were incubated in a 37 °C incubator for 12 h. The bactericidal rate, E_b (three duplicated samples' average value) was calculated by the following equation:

$$E_b = \frac{N_c - N_s}{N_c}$$

where N_c and N_s are the quantity of colonies associated with to glass and AFC-*x*, respectively.

2.4. Anti-protein adhesion test

BSA-FITC, a model protein tagged with fluorescein isothiocyanate, was used to assess the coating's capacity to bind to proteins. The BSA-FITC solution was diluted with PBS buffer to 0.1 mg/mL. Prior to testing, the AFC-*x* coatings were introduced into 10 mL of FITC-BSA solution at a concentration of 0.1 mg/mL in a six-well plate and soaked under dark conditions for 24 h. Next, the surfaces of the AFC-*x* coatings were gently washed with PBS solution to remove the unattached protein molecules, and dried for 10 min. Finally, the surfaces of the coatings were observed under a fluorescence microscope (MF52-N) to evaluate their anti-protein adhesion properties. The magnification of the fluorescence microscope was set at 20 times, and the fluorescein isothiocyanate produced yellow-green fluorescence under blue light excitation.

2.5. Anti-algal adhesion test

Diatom was selected as algal contaminant to test the anti-algal adhesion of the AFC-*x* coatings. Initially, 1 g of the diatom species and

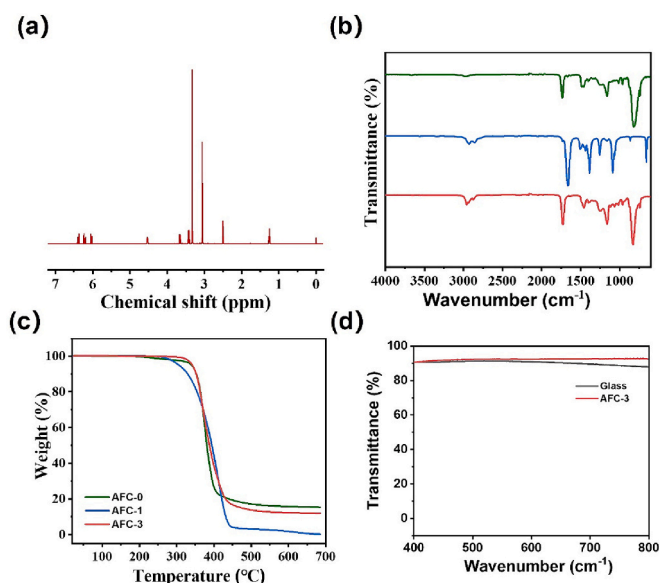


Fig. 2. (a) ^1H NMR spectra of IL- PF_6 and (b) FTIR spectra of AFC-0, AFC-1 and AFC-3 coatings. (c) TGA spectra of AFC-0, AFC-1 and AFC-3 coatings. (d) Transmittance of glass and AFC-3 coating.

2–3 drops of diatom culture liquid were poured into 1000 g of artificial seawater in a glass sink and nurtured for 7 days. Then, the AFC-x coatings were taken into the glass sink and cultivated for 7 days. Finally, the AFC-x coatings were gently washed with deionized water and observed under an optical microscope (MF52-N). The magnification of the fluorescence microscope was operated at 20 times.

2.6. Self-cleaning test

Firstly, hydrophilic copper chloride ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) particles and hydrophobic silicon dioxide (SiO_2) powder were respectively sprayed on the surface of the AFC-x coatings. Then the AFC-x coatings were placed at an inclined position, followed by the addition of water droplets to their surfaces to remove surface solid contaminants, and the cleaning effect of the water droplets was observed to depict the self-cleaning property of the AFC-x coatings.

3. Results and discussion

3.1. Structure and characterization of the AFC-x coatings

The chemical shift peaks at 6.40 ppm and 6.36 ppm are suggestive of 2 protons on $-\text{CH}=\text{CH}_2$, the peak at 6.04 ppm relates to 1 proton on $-\text{CH}=\text{CH}_2$, and the peak at 4.53 ppm correspond to 2 protons on $-\text{O}-\text{CH}_2-\text{CH}_2$. The peak at 3.66 ppm relates to 2 protons on $-\text{N}-\text{CH}_2-\text{CH}_2$, while the peaks at 3.42 ppm, 3.06 ppm, and 1.26 ppm are suggestive of 2 protons on $-\text{H}-\text{CH}_2-\text{CH}_2$, 6 protons on $-\text{N}-(\text{CH}_3)_2$, and 3 protons on $-\text{CH}_2-\text{CH}_3$ (Fig. 2a).

The chemistry of compositions of the AFC-x was characterized by FTIR, SEM, EDS and TG-DTG spectra. From Fig. 2b, the absorption peaks of AFC-x at 2967 – 2854 cm^{-1} , 1730 cm^{-1} , 1486 cm^{-1} and 1160 cm^{-1} are ascribed to the stretching vibration of $\text{C}-\text{H}$, $\text{C}=\text{O}$, $-\text{CH}_2-$ and $\text{C}-\text{N}$ [23]. Besides, the bending vibration of $\text{P}-\text{F}$ at 825 cm^{-1} appeared between the AFC-0 and AFC-3 coatings [24], but was not characterized in the AFC-1 coating.

The characteristics of the acrylic coating of “hot sticking and cold shrinkage” showed that the acrylic coatings exhibited poor heat resistance. The initial decomposition temperature of AFC-1 coating was 269 $^\circ\text{C}$, while that of AFC-3 coating was 316 $^\circ\text{C}$ (Figs. 2c and S1). These indicate that the heat resistance of the acrylic coatings were improved

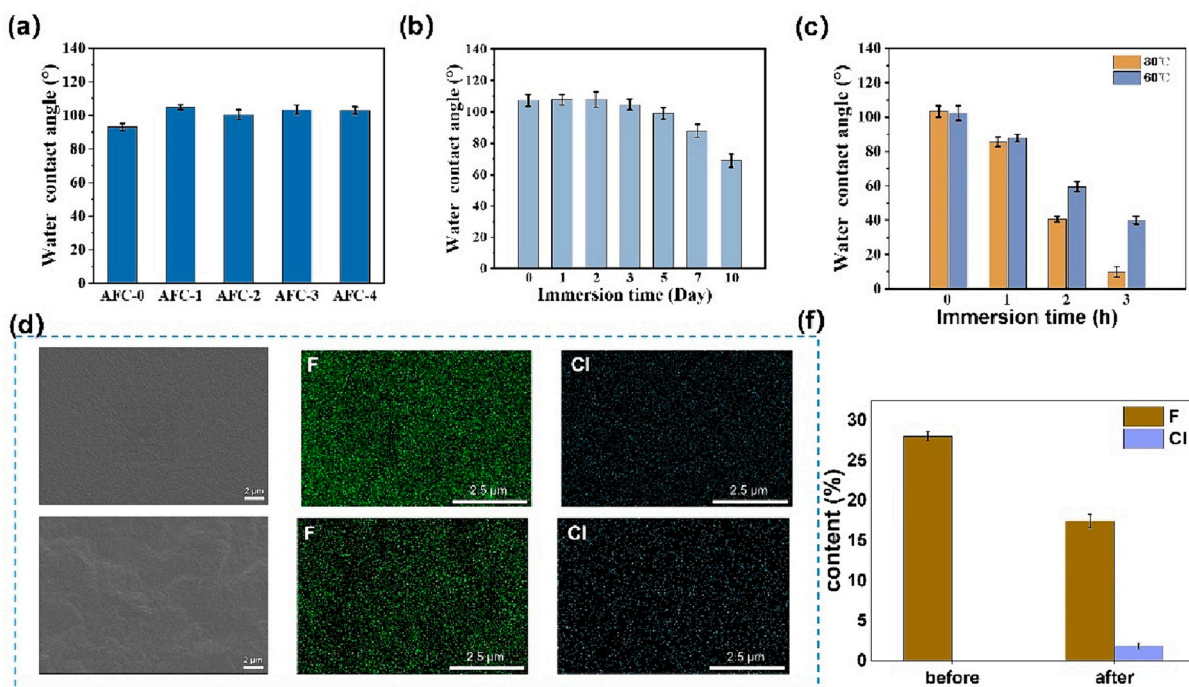


Fig. 3. (a) The WCA of the AFC-x coatings. The WCA of the AFC-3 coating after 10 days submersion in artificial seawater at room temperature (b) and at 60 $^\circ\text{C}$ and 80 $^\circ\text{C}$ for 3 h (c) SEM images of the AFC-3 coating before (d) and after (e) submersion in artificial seawater for 3 days. (f) The atomic distribution of F and Cl on the AFC-3 coating before and after being submerged in artificial seawater for 3 days.

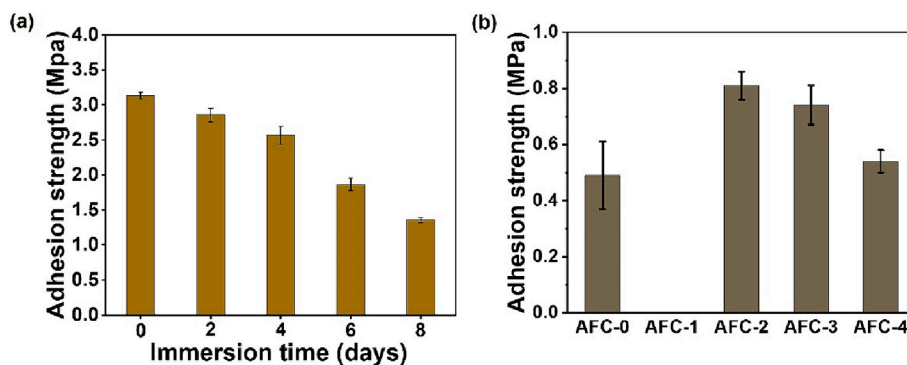


Fig. 4. The adhesion of the AFC-3 coating evaluated by (a) lap shear test during the immersion and (b) the adhesion of coating evaluated by 90° peeling test.

with the introduction of PF_6^- , thus extending the application field of the AFC-3 coating [25,26]. The cross-section was smooth (Fig. S2), flat and without obvious folds and defects, and no phase separation was observed, indicating that the AFC-3 coating had good film formation.

The transmittance of the AFC-3 coating was over 92 % between 400 and 800 cm^{-1} (Fig. 2d), which is consistent with the FESEM analysis, proving its application in optical devices [27], as well as protective films on material surfaces that need to remain clear and transparent.

3.2. Surface wettability of the AFC-x coatings

The (WCA) of the AFC-x coatings reflected their surface wettability. The WCA of all the samples were above 90°, indicating good hydrophobicity. While AFC-0 coating exhibited the least WCA, the highest WCA was recorded for AFC-1 coating. Moreover, the WCA of AFC-2, AFC-3 and AFC-4 first increased and then decreased with increasing content of IL- PF_6 in the ionic liquid monomer (Fig. 2a). The phenomenon above illustrates that the long chain alkane and ionic liquid affected the wettability of the coatings to some extent.

The hydrophobic anion (PF_6^-) contained in the AFC-3 coating was converted to the hydrophilic anion (Cl^-) by ion exchange, to afford hydrophilic properties on the coating. As shown in Fig. 2b, the WCA of the AFC-3 coating was initially 107°, but it decreased to 103.358° following 3 days spent submerged in artificial seawater, illustrating the onset of Cl^- and PF_6^- exchange on the surface. The results are consistent with those of SEM and EDS following 3 days spent submerged in artificial seawater. After 60 days immersion, the WCA of the AFC-3 coating became 11.356 °C, demonstrating the change in its surface property from hydrophobic to hydrophilic. Furthermore, the AFC-3 coating was submerged in artificial seawater at 60 °C and 80 °C to explore the influence of heating conditions on the ion exchange rate (Fig. 2c). The drop rate of the WCA test was faster at 80 °C compared with the test at 60 °C. The WCA of the AFC-3 coating was 10.081 °C after being submerged in artificial seawater at 80 °C for 3 h. This displays the complete transition of the AFC-3 coating surface from hydrophobic to super-hydrophilic. From the reported observation, it is clear that high temperature accelerated the ion exchange reaction and the static effectiveness of the AFC-3 coating in artificial seawater at room temperature.

According to the SEM micrograph of the AFC-3 coating (Fig. 2d), the surface was smooth without obvious folds, convincingly demonstrating the successful synthesis of the AFC-3 coating with good film formation. In addition, there was a slight wrinkle on the surface of the AFC-3 coating (Fig. 2f) after 3 days immersion in artificial seawater, illustrating the exchange between Cl^- and PF_6^- . The fluorine distribution on the surface of the AFC-x coating was more intense before immersion in artificial seawater compared with the observation after immersion, but chlorine manifested a contrasting behavior with the post-immersion exhibiting a more intense distribution (Figs. 2f and S3). From Table S2, it is obvious that the Cl/F proportion on the coating's surface after being submerged in artificial seawater was much higher than that

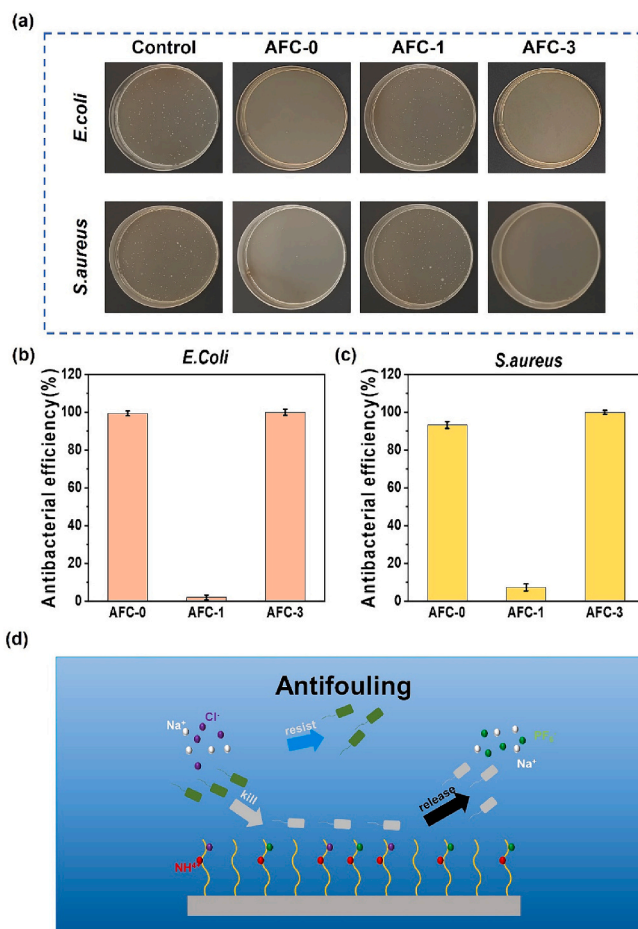


Fig. 5. Anti-*E.coli* and anti-*S.aureus* adhesion accomplishments (a) and the antibacterial efficiency regarding (b) *E. coli* and (c) *S. aureus* of blank control and of the AFC-x coatings. (d) Antifouling mechanism of the AFC-3 coating.

before immersion. The elemental distribution further described the exchange between Cl^- and PF_6^- , and is consistent with the pre- and post-immersion SEM micrographs of the AFC-x coating. These results prove that the coating is capable of dynamic surface responsiveness with switchable wettability properties. (See Fig. 3.)

3.3. Mechanical properties and adhesion to substrate of the AFC-x coatings

Due to their poor toughness, AFC-0 and AFC-4 coatings were not subjected to tensile tests. Also, tensile tests were not performed on AFC-1

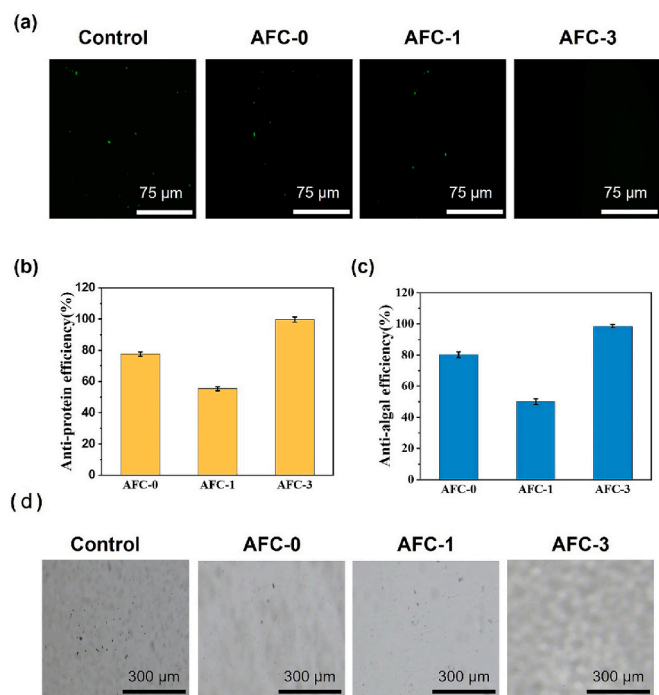


Fig. 6. Anti-protein accomplishments (a) of the AFC-x coatings. Anti-protein (b) and anti-algal (c) efficiency of the AFC-x coatings. Anti-algal accomplishments (d) of the AFC-x coatings.

coating due to its relatively poor tensile strength. Thus, only for AFC-2 and AFC-3 coatings were tested for their tensile properties. The tensile strength of AFC-3 coating reached 14 MPa, which is almost twice the value for AFC-2 coating, and its elastic modulus reached 331.43 MPa, which is 200 times that of AFC-2 coating (Table S3). These reveal that AFC-3 coating showed excellent tensile properties in the uniaxial tensile tests.

The adhesive force of AFC-1 coating on the substrate was too small to be effectively tested by the lap shear test and 90° peel test. As shown in Fig. 4a-b, the adhesion force of the AFC-3 coating was 3.135 MPa by lap shear test and 0.74 MPa by 90° peel test, which meet the requirements

for adhesion strength between coatings and substrates in real applications. After 8 days immersion, the adhesion of AFC-3 coating became 1.356 MPa, which still meet the requirements for adhesion strength between coatings and substrates in real applications.

3.4. Anti-fouling performance property of the AFC-x coatings

It is well known that surface QACs exhibit excellent biocidal activities, where the positive charges on the QAC group attract negatively charged bacterial cells [28,29], however, the antibacterial effectiveness of the coatings is diminished in consequence of the build-up of too many dead germs on the coatings' surface. Therefore, the dynamic surface formed by the ionic exchange of PF_6^- and Cl^- contributed to fouling release and improved the hydrophilicity of the coating, thereby inhibiting protein adsorption. As shown in Fig. 5a-c, Regarding *E. coli* and *S. aureus*, the AFC-3 coating exhibited anti-bacterial effectiveness of 100 % and 99.96 %, severally, and these are ascribed to the synergistic effect of the QACs and dynamic surface [30–32], and fouling is release from PF_6^- and Cl^- exchange (Fig. 5d). Besides, the AFC-3 coating exhibited excellent anti-protein property (Fig. 6a), due to the enhanced hydrophilicity [4] of the coating by the ion exchange between Cl^- and PF_6^- . Moreover, compared with glass, there was no obvious diatom attachment on the AFC-3 surface (Fig. 6d). The above results show that the coating had excellent algal resistance because of the synergistic effect of the inhibition of algal attachment on the QACs and the enhanced hydrophilicity from PF_6^- and Cl^- exchange.

The AFC-3 coating exhibited outstanding anti-fouling properties. This is mainly attributed to the combined actions of the QACs, which killed the existing microorganisms without causing potential risks to environment, and the ionic exchange of the Cl^- and PF_6^- , which was beneficial to the formation of a dynamic surface for fouling release and enhanced hydrophilicity of the coating. Besides, the hydrophilic coating surface favored the inhibition of protein adhesion. With continuous PF_6^- - Cl^- exchange, the surface hydrophilicity of the AFC-3 coating was increasingly enhanced, which benefitted the inhibition of protein adhesion.

3.5. Self-cleaning property of the AFC-3 coating

As shown in Fig. 7a-b, the AFC-3 coating had been treated with

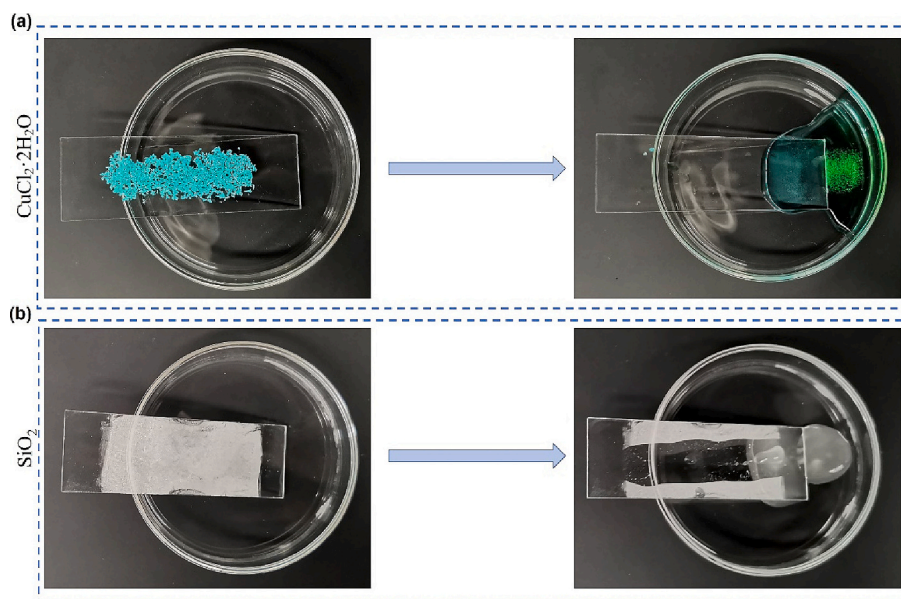


Fig. 7. The Self-cleaning performance to (a) hydrophilic copper chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) particle contaminants and (b) Hydrophobic silicon dioxide (SiO_2) powder contaminants.

pollutants, and the deionized water droplets were delivered at an angle of roughly 13° above the contaminants. The hydrophilic contaminants were dissolved and rolled off the AFC-3 coating surface. The hydrophobic contaminants were dispersed by the water droplets, with the exception for extremely little SiO₂ residues on the AFC-3 coating surface. As a result, the AFC-3 coating's self-cleaning capability was crucial in eliminating various solid impurities in real-world applications.

4. Conclusions

In summary, we developed an enhanced antifouling strategy for switching between self-renewing and hydrophilic surfaces fabricated by the exchange of PF₆[−] and Cl[−] in artificial seawater and contact bactericidal mechanisms. The self-renewing surface contributing to fouling release and the formation of hydrated layers is also conducive for the resistance of protein adhesion. The AFC-3 coating was prepared by the free radical photo-copolymerization of 2-EHA, BA and an ionic liquid monomer IL-PF₆ obtained via ion exchange reaction. The anti-bacterial efficiency of the AFC-3 coating was 100 % against *E. coli* and 99.96 % against *S. aureus*, respectively. After 60 days immersion in artificial seawater at room temperature, the WCA of the AFC-3 coating became 11.356 °C, enhancing the inhibition of protein adsorption. Therefore, the AFC-3 coating is highly promising for a broader range of marine antifouling and biomedical materials. Moreover, the transmittance of the AFC-3 coating was over 92 % between 400 and 800 cm^{−1}. Thus, it also has potentials for applications on the surface of optical devices, such as underwater cameras.

CRediT authorship contribution statement

Qiang Li: Writing – original draft. **Qi Yang:** Writing – original draft. **Zheng Cao:** Methodology. **Xiang Le:** Conceptualization. **Fandong Meng:** Writing – review & editing. **Yongyuan Ren:** Writing – review & editing. **Quan Liu:** Writing – review & editing. **Qinghua Zhang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2024.108471>.

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